Artificial Intelligence Pipeline for Scalable, Efficient Design of Next-Gen Wireless Circuits

Jolie Wang[1], Matthew Karazincir[1], Chris Han[1], **Rice Integrated Systems & Electromagnetics Lab[1]**
[1]Department of Electrical and Computer Engineering, Rice University

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Overhead of Wireless Circuit Design

Revolutionary AI + High Costs of Wireless Circuit Design

- Wireless circuits are a crucial component of every wireless link, radar system, and sensing platform, from 5G handsets to autonomous vehicles.
- **The cost of developing these components is substantial:**
  - Technical expertise is needed at every step
  - Simulations take hours to even days
- Though AI is revolutionizing and streamlining other essential workflows, **its application to wireless circuit design faces unique challenges:**
  - Underlying dependence on circuit physics; **nonlinear behavior**
  - High reliance on **expert human intuition**

AI to Accelerate Design

- Design a user-friendly large language model (LLM) interface, answering wireless design questions with **insightful technical detail**
- Leverage state-of-the-art AI techniques and models for **accelerating wireless chip design (including optimization)**

Design Criteria

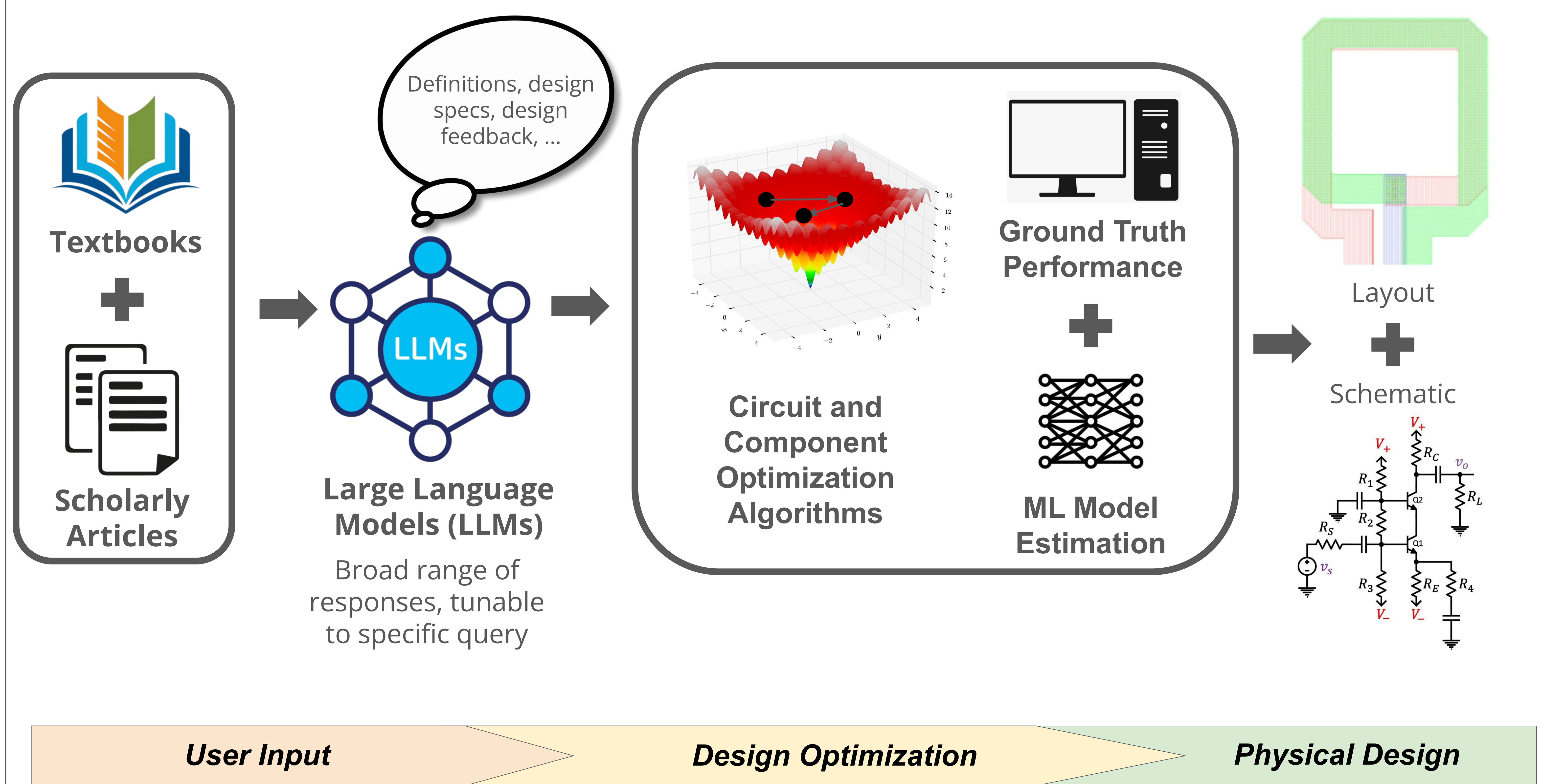
User Input
- A chatbot-based user-friendly interface
- Efficient answers
  - References to textbooks & journals

Design Optimization
- Integrates AI to speed up design time
- Handles increasingly complex designs
  - Topology, input data type, size
- Effective across all stages of design
  - Amplif., power delivery, etc.

Physical Design
- Synthesize functional circuits and components
- Enable rapid prototyping and validation
  - Reduced time spent at bottlenecks

Scalable + Efficient + Broadly Design-Applicable

AI-Integrated Workflow



AI-Enabled Wireless Design Improvements

Improved LLM Accuracy
- Baseline: 88.67%
- Our LLM: 91.00% **(+2.33% accuracy)**

Example Human-LLM Interaction:
- Question: "What is a cascode PA?"
- Answer: "A cascode power amplifier (PA) is a configuration that typically involves two transistors arranged in a stacked manner..."

Lower EM Simulation Model Error
- Previous works: 2.10%
- Our Model: 1.29% **(38.57% improvement)**

Designs Robust Across:
- **Device templates**
  - Transformer
  - T-Coil
  - Inductor
- **Fabrication Processes**
  - CMOS SOI
  - InP
  - GaN

Faster, more reliable optimization
- Significant convergence: **52.2% → 91%**
- Median optimization time: **36.7min → 8.3min**

Scalable + Adaptive ML usage:
- Focuses optimization **in most promising directions** over time, negating high dim.
- **Adapts percent of runs on ML** per step based on its current performance score

Next Steps

- Creating a visual interface that shows each step's progress live to user
- Improving LLM accuracy with models trained specifically on wireless design knowledge
- Co-optimization of passive (e.g. coil inductor) and active components (e.g. power amplifier)

Acknowledgements

We would like to thank our mentor, Houbo He, and our faculty sponsor, Dr. Taiyun Chi, for their mentorship and support throughout the academic year. We would also like to thank Dr. Gary Woods and the rest of the RISE lab for their assistance.